

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	Members need to be able to search the ISBN repository and, if the book is already checked out, drop a hold on it and get slotted into a first-in-first-out waitlist. Period.	High	Pass: The user lands in the correct FIFO position every time. Fail: Someone who's already at the maximum allowed number of books still manages to place yet another reservation. That's not okay.	Reserving books that aren't currently available is basic, core library behavior.	
FR-002	Functional	Student Members should be able to look up a book in the library catalogue and immediately see what it is and whether it's actually available. No guessing.	High	Pass: Enter a valid ISBN and the system shows the title, author, ISBN, plus the current availability status. Fail: Any of those required details are missing, wrong, or inconsistent.	Students won't reserve what they can't confidently identify.	
FR-003	Functional	When a reserved book finally becomes available, the system needs to ping the next Student Member in line automatically. No manual chasing.	High	Pass: The next eligible person in the hold queue receives a notification right when the book becomes available. Fail: No notification goes out, or it goes to the wrong person in the queue. That defeats the whole point.	People should know the moment their reserved book is ready.	

FR-004	Functional	The Head Librarian must be able to manage holding queues and adjust reservation states for books. They're in charge, they need the tools.	Medium	Pass: The Head Librarian can view queue entries, update them, remove them, and change a book's reservation status without roadblocks. Fail: Any queue or status action that an authorized Head Librarian should be able to do can't be completed.	Reservation workflows need administrative control, or they fall apart fast.	
FR-005	Functional	The system has to log overdue fine transactions and let the Head Librarian track them per member. This is money-related. It can't be fuzzy.	Medium	Pass: Each fine is stored with the correct member, the amount, and the transaction status, and it can be retrieved reliably. Fail: The system can't record it properly or can't pull it back accurately later.	Overdue fines are financial transactions, and tracking them is mandatory.	
NFR-001	Nonfunctional	The catalog search query shall return paginated results for 50,000 titles in less than 150 ms.	High	Pass: Benchmark testing shows the query returns results in under 150 ms under the specified load. Fail: Response time is 150 ms or greater.	Ensures fast catalogue searching even with a large number of titles.	
NFR-002	Nonfunctional	The system shall protect library member and fine transaction data from unauthorized access.	High	Pass: Unauthorized users cannot access member or fine information during security testing. Fail: An unauthorized user can view or modify protected data.	Protects sensitive library and transaction information.	

